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Module 4 Introduction

Lesson 1: Normal Approximation and Its Applications

Lesson 2: Exploring Large-Sample Theory and Counterexamples

- **Reading:** Exploring Large-Sample Theory and Counterexamples Readings 2h
- **Video:** Large-Sample Theory Pt. 1 7 min
- **Video:** Large-Sample Theory Pt. 2 9 min
- **Video:** Large-Sample Theory Pt. 3 5 min
- **Video:** Large-Sample Theory Pt. 4 8 min
- **Video:** Large-Sample Theory Pt. 5 7 min
- **Video:** Large-Sample Theory Pt. 6 6 min
- **Practice Assignment:** Exploring Large-Sample Theory and Counterexamples Quiz Submitted

Lesson 3: Frequency Properties and Broader Interpretations of Bayesian Methods

Module 4 Summative Assessment

Module 4 Summary

Your grade: 100%

Your latest: **100%** • Your highest score: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

Exploring Large-Sample Theory and Counterexamples Quiz

Instructions

Review Learning Objectives

1. What does asymptotic theory describe in Bayesian analysis? **1 / 1 point**

   - Prior distributions for small samples
   - Properties of Bayesian models as sample size approaches infinity
   - Behavior of posterior distributions as sample size becomes large
   - Exact posterior distributions

   Correct
   Correct! Asymptotic theory focuses on large-sample properties.

2. What does the Fisher information measure in the context of Bayesian analysis? **1 / 1 point**

   - Prior distribution accuracy
   - Information provided by the data
   - Sampling variability
   - Posterior distribution precision

   Correct
   Fisher information quantifies the amount of information the data provide about a parameter.

3. What is the impact of increasing sample size on the influence of the prior distribution in Bayesian analysis? **1 / 1 point**

   - The influence of the prior increases
   - The influence of the prior decreases
   - The influence becomes negligible only if the prior is informative
   - The influence remains the same

   Correct
   Correct! The posterior is a balance between the prior and the data. As the data grows larger, however, it dominates.

Assignment details

Submitted	Attempts	
Nov 22, 1:53 AM IST	Unlimited	Retry

Your grade
To pass you need at least 80%. We keep your highest score.
100%

View submission | See feedback